# Exam - 1 Review

*Concepts, must-know procedures, mini-examples, and common traps*

(Scope: evaluating expressions & like terms; one-variable equations & inequalities; systems by substitution & elimination; relations, slope, line forms; graphing lines & linear inequalities)

**How to Use These Notes**

**Model first, compute second.** Sketch a tiny diagram/table, mark what is given, pick the right form, then substitute carefully.
Keep exact values until the end; round only in the final answer if asked.

## 1. Evaluating Expressions & Like Terms

**Core Ideas**

**Evaluate by substitution:** replace variables, then follow order of operations (exponents $\to$ multiply/divide $\to$ add/subtract).
**Like terms:** same variable(s) with the same exponents combine by adding coefficients.

**Mini-examples.**

a) For $x = -1$, $y = 2$: $(2x - 3y)^2 = \big(2(-1) - 3(2)\big)^2 = (-8)^2 = 64$.

b) Simplify, then evaluate at $p = 3$: $2p^2 + 3p - p^2 + 7 - 4p + 2p^2 \to (2-1+2)p^2 + (3-4)p + 7 = 3p^2 - p + 7 \Rightarrow 3(9) - 3 + 7 = 31$.

**Helpful tips.**

- Use parentheses for negatives: $(-3)^2 = 9$ but $-3^2 = -9$.
- Do not distribute exponents across sums: $(a + b)^2 \neq a^2 + b^2$.
- Combine like terms before substituting numbers to reduce errors.

## 2. Linear Equations & Inequalities (one variable)

**Solve-It Playbook**

**Equations:** Distribute $\to$ combine like terms $\to$ move variables to one side $\to$ undo $\times/\div$.
**Inequalities:** Same steps, but <u>flip the sign</u> when multiplying/dividing both sides by a negative.
**Quick check:** plug your solution back into the original.

**Mini-examples.**

a) Solve $\frac{3}{4}(x-2)-\frac{1}{2}(x+6)=5$.
$\frac{3}{4}x-\frac{3}{2}-\frac{1}{2}x-3=5 \Rightarrow \left(\frac{3}{4}-\frac{1}{2}\right)x-4.5=5 \Rightarrow \frac{1}{4}x=9.5 \Rightarrow x=38$.

b) Solve and mark the flip: $-\frac{1}{2}(4x+6)\geq 7 \Rightarrow -2x-3\geq 7 \Rightarrow -2x\geq 10 \Rightarrow x\leq -5$.

**Common traps.**

- Forgetting to flip $<,>,\leq,\geq$ after dividing/multiplying by a negative.
- Dropping a term when combining across the sign.

# 3. Systems of Linear Equations & Intersections

Big Idea

The graph of a line is the set of points making its equation true. The intersection point lies on both lines, so it satisfies both equations simultaneously $\Rightarrow$ it is the solution. **Outcomes:** one solution (lines cross once), no solution (parallel distinct), infinitely many (same line).

**Pick a method.**

- **Substitution:** when a variable is easy to isolate.
- **Elimination:** when coefficients line up (or can be made to).

**Mini-examples.**

a) Eliminate: $\begin{cases} 4x-7y=-1 \\ 6x+7y=41 \end{cases} \Rightarrow 10x=40 \Rightarrow x=4,\ y=\frac{17}{7}$.

b) Special outcomes after eliminating: $0=0$ (infinitely many solutions); $0=5$ (no solution).

**Helpful tips.**

- Convert both to $y=mx+b$ to compare slopes/intercepts quickly.
- Check the found point in *both* equations.

# 4. Relations, Slope, and Line Forms

Must-Know

**Relation:** set of ordered pairs $(x,y)$. **Domain:** all $x$. **Range:** all $y$.
**Slope:** $m=\dfrac{y_2-y_1}{x_2-x_1}$ (rate of change). **Slope-intercept:** $y=mx+b$. **Point-slope:** $y-y_1=m(x-x_1)$.
Horizontal line: $y=b$ (slope 0). Vertical line: $x=a$ (undefined slope).

**Mini-examples.**

a) Two points to $y = mx + b$: $(-1, 2)$ and $(5, -4) \Rightarrow m = \frac{-6}{6} = -1$, so $y = -x + 1$.

b) From slope and a point: slope 3 through $(2, -5)$ gives $y + 5 = 3(x - 2) \Rightarrow y = 3x - 11$.

**Helpful tips.**

- Write point-slope first, then expand if needed.
- For tables, constant first differences in $y$ indicate linear.
- $x$-intercept from $y = mx + b$: set $y = 0$, so $x = -\dfrac{b}{m}$ (if $m \neq 0$).

# 5. Graphing Lines & Linear Inequalities

**Shade Like a Pro**

**Boundary:** replace the inequality with its line. **Solid** if $\leq$ or $\geq$, **dashed** if $<$ or $>$.
**Which side?** Test a convenient point (often $(0, 0)$). If it satisfies the inequality, shade that side.
**Systems of inequalities:** the solution is the overlap of the shaded regions.

**Mini-examples.**

a) Point vs line: Is $(2, -1)$ above or below $y = -3x + 4$? At $x = 2$, the line has $y = -2$. Since $-1 > -2$, the point is above.

b) Style: $y > 3x - 2$ is dashed; $y \geq -x + 6$ is solid.

c) Parallel strip between $y = 2x + 1$ and $y = 2x - 5$: write $2x - 5 \leq y \leq 2x + 1$.

**Helpful tips.**

- Rewrite as $y = mx + b$ before "above/below" checks.
- A single valid point is enough for "find any point that works."
- Equal slopes and different intercepts $\Rightarrow$ no solution (parallel lines).

# 6. Quick Mixed Practice (self-check)

Try without a calculator; answers below.

1) Evaluate at $x = 3$: $2x^2 - 5x + 4$.

2) Simplify, then evaluate at $p = 3$: $2p^2 + 3p - p^2 + 7 - 4p + 2p^2$.

3) Solve: $\dfrac{3}{4}(x - 2) - \dfrac{1}{2}(x + 6) = 5$.

4) Solve: $-\frac{1}{2}(4x+6) \geq 7$ (mark where you flip).

5) Classify/solve: $y = 3x + 1$ and $6x - 2y = 10$.

6) Eliminate quickly: $\begin{cases} 4x - 7y = -1 \\ 6x + 7y = 41 \end{cases}$.

7) Slope & form: through $(2, -5)$ with slope 3.

8) Two points to $y = mx + b$: $(-1, 2)$ and $(5, -4)$.

9) Which side is $(2, -1)$ relative to $y = -3x + 4$?

10) Write the "strip" between $y = 2x + 1$ and $y = 2x - 5$.

11) Inequality styles: dashed/solid for $y > 3x - 2$ and $y \geq -x + 6$.

12) Give one point satisfying $y < 2x - 1$ and $y \geq -x + 3$.

**Answers (peek after you try)**

1) 7. 2) 31. 3) $x = 38$. 4) $x \leq -5$. 5) Parallel $\Rightarrow$ no solution. 6) $(x, y) = \left(4, \frac{17}{7}\right)$. 7) $y = 3x - 11$. 8) $y = -x + 1$. 9) Above (line value $-2$, point $-1 > -2$). 10) $2x - 5 \leq y \leq 2x + 1$. 11) $y > 3x - 2$ dashed; $y \geq -x + 6$ solid. 12) Example: $(2, 2)$.

*Final checklist:* (i) Pick a form that minimizes algebra; (ii) keep both sides balanced; (iii) flip the sign when dividing by a negative in inequalities; (iv) justify why an intersection solves a system; (v) mark solid vs. dashed boundaries correctly.